Ex 13)

Implement your own ping program

Aim:

To implement a ping program using python raw sockets to check the reachability of a host and calculate the round trip time.

Code:

```
import socket
import struct
import time
import os

def checksum(data):
    total = 0
    for i in range(0, len(data)-1, 2):
        total += (data[i] << 8) + data[i+
    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xFFF
    total += total >> 16
    return (~total) & 0xFFFF

def ping(host):
    try:
        dest_addr = socket.gethostbyname(hos
        sock = socket.socket(socket.AF_IN
                             socket.SOCK_RAW,
                             socket.IPPROTO_ICMP
```

```
packet_id = os.getpid() & 0xFFFF
Sequence = 1
header = Struct.pack("!BBHHH", 8,
                     0, 0, packet_id,
                     Sequence)
data = b"PING"
header = Struct.pack("!BBHHH", 8,
                     0, checksum, packet_id,
                     Sequence)
Packet = header + data
start = time.time()
end = time.time()
rtt = (end - start) * 1000
sock.close()
except Socket.timeout:
        print("request time out")
except Permission Error:
print("Permission denied. Run the
                program as admin!)
ping(host)
```

Sample output:

Enter destination host : 8.8.8.8

Reply from 8.8.8.8 : time = 24.53 ms

Result:

Thus, the ping program was successfully implemented in Python using raw Sockets.